Docker Compose

Recall:

- **Docker image** = is a read-only blueprint or template used to create containers.
  - Think of it like a recipe for making a cake
- **Docker Container** = is a running instance of an image

`docker network` = Allows containers to communicated with each other and with the external world.

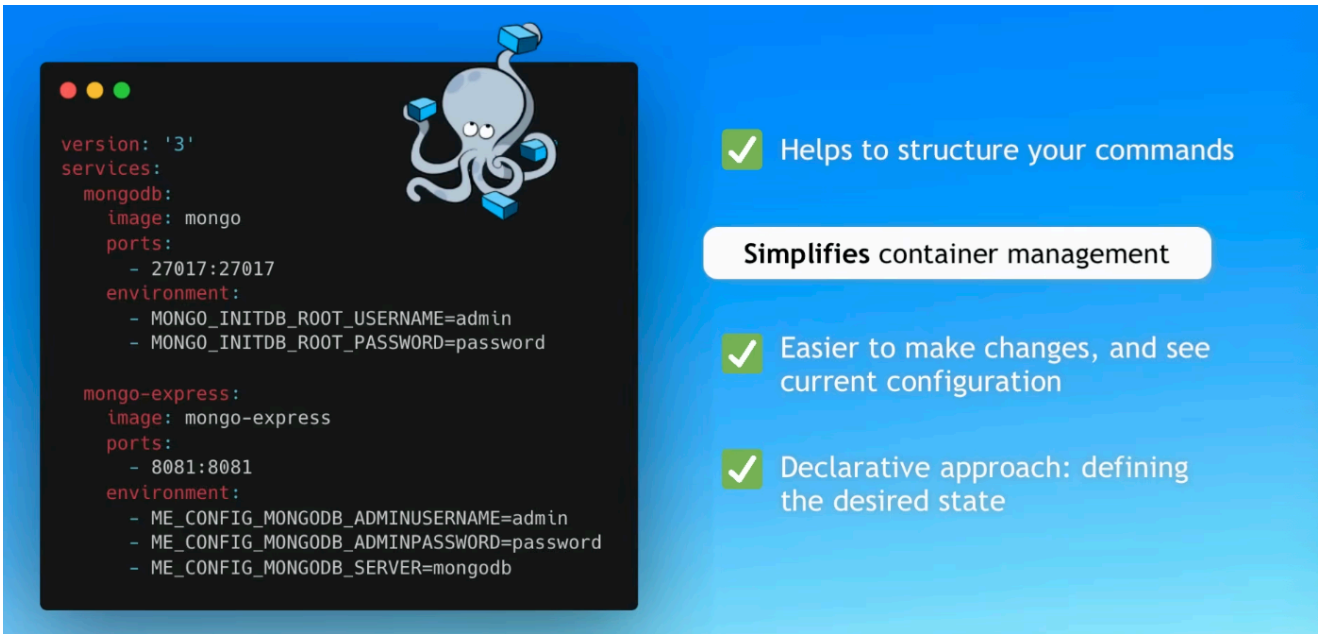
`docker network create {name}` = to create a new user-defined network.

`docker network ls`

```
docker run -d \
-p 27017:27017 \
-e MONGO_INITDB_ROOT_USERNAME=admin \
-e MONGO_INITDB_ROOT_PASSWORD=supersecret \
--network mongo-network \
--name mongodb \
mongo
```

Option	Meaning
`docker run`	Start a container
`-d`	Run in background (detached mode)
`-p 27017:27017`	Expose MongoDB port
`-e`	Set environment variables
`--network mongo-network`	Attach to Docker network
`--name mongodb`	Container name
`mongo`	MongoDB image

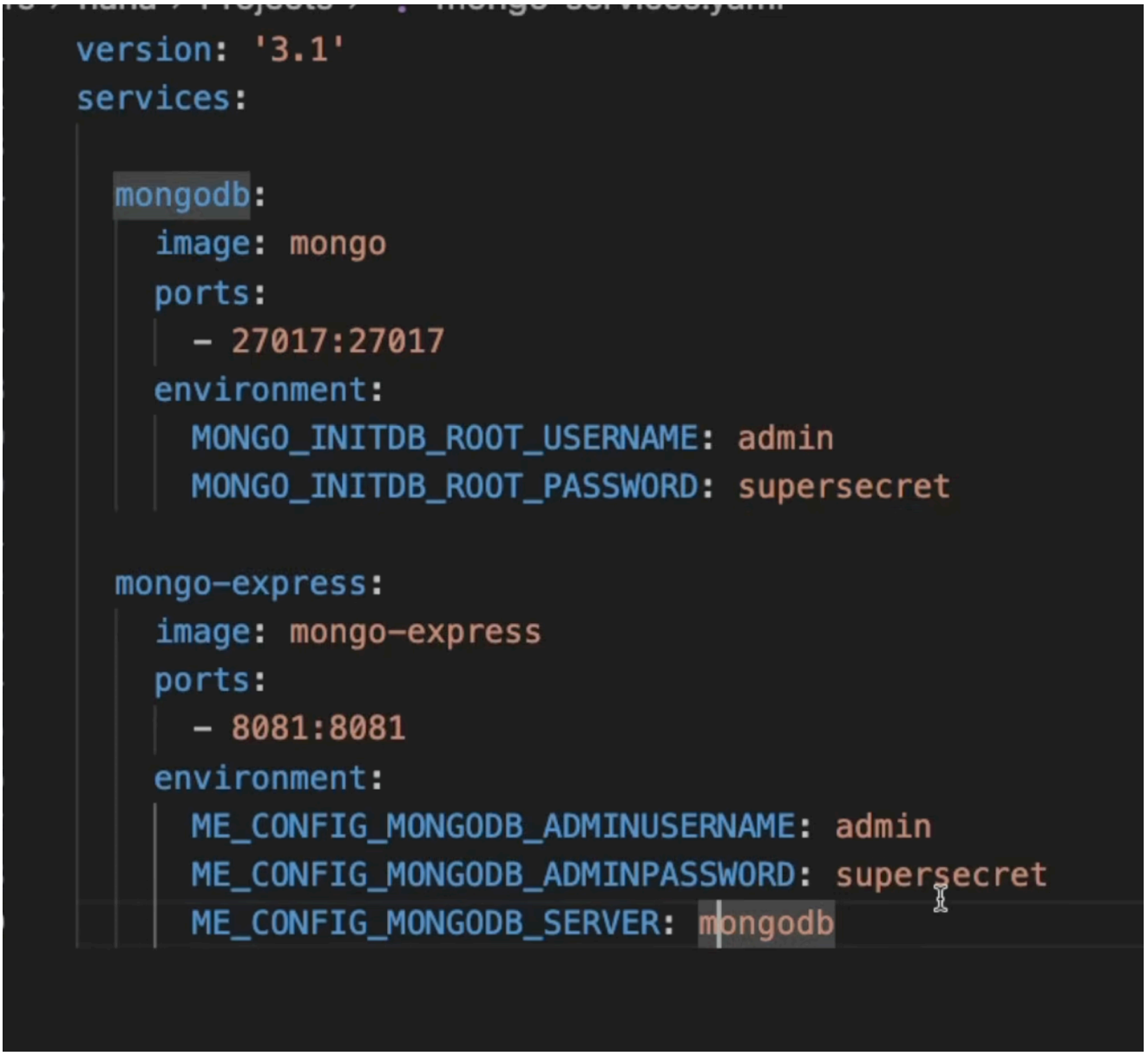
Here's where the docker compose becomes handy



" "X" as code " concept:

- Code that defines how your services should run
- Code can be versioned
- Easier collaboration

One more example of docker compose



`docker-compose -f {name}.yaml up`
`docker-compose -f {name}.yaml down` = Stops containers and removes containers, networks, volumes and image created by 'up'

`docker ps` = list running containers on the system

`docker ps -a` = show all containers

`docker container ls` = list your running containers.

`docker container ls -a` = list all the running containers including those that are stopped

Instead of using the `docker-compose -f {name}.yaml down`, we can temporarily stop the compose by `docker-compose -f {name}.yaml stop` and resume it by calling `docker-compose -f {name}.yaml start`

- Stops the running containers
  - Keeps:
    - containers

- networks
- volumes
- container state

Rename the project
`docker-compose --project-name {name} -f {name}.yaml up`

Main use case of Docker Compose

- To define and manage services (containers) that make up your app
- So they can be run together in an isolated environment
- Easy to run and clean up your entire app

Limitations

- Well-suited for local development and small-scale deployment
- Designed to run containers on a single host system
- Engineers still need to operate the containers manually.

  Kubernetes to the rescue\

To build an image

`docker build -t {name}`

docker compose pull` = pull the images

Push the compose to the docker hub
`docker compose build`

Authenticate
`docker login`

Push it to docker hub
`docker compose push`